

Online Quiz Management System

Programming in Java - Project Report

1. Introduction

The Online Quiz Management System is a Java command-line application developed to simplify basic quiz administration. It provides separate student and administrator functions and stores application data in local files.

2. Problem Statement

Manual quiz management requires effort for question distribution, checking answers, calculating scores, and maintaining result records. The proposed system automates these basic activities in a simple terminal-based environment.

3. Objectives

Provide student registration and authentication; allow students to attempt a quiz; calculate scores automatically; store and display results; allow an administrator to manage questions; demonstrate core Java programming concepts.

4. Functional Requirements

User Management, Quiz Management, and Quiz and Result Management. Students can register, log in, attempt quizzes, and view results. Administrators can manage questions and view results.

5. Non-Functional Requirements

Performance, usability, reliability, maintainability, error handling, and resource efficiency.

6. System Architecture

The CLI communicates with Main.java. Main controls navigation and calls the student, administrator, quiz, result, and file-management classes. FileManager handles local data storage.

7. Design Decisions and Rationale

Java was selected because the project is for Programming in Java. A command-line interface keeps the project executable in a terminal without requiring GUI configuration. Text files provide simple persistent storage without an external database dependency. Separate classes keep responsibilities modular.

8. Implementation Details

The application uses an abstract User class with Student and Admin subclasses. Question represents an individual question, Quiz manages a collection of questions and calculates scores, Result represents a submitted result, QuizManager manages questions, and FileManager handles persistence.

9. Testing Approach

QuizTest.java verifies that a quiz containing known correct answers returns the expected score. Manual tests should cover registration, login, invalid input, quiz submission, result display, and question management.

10. Results / Demonstration

The demonstration should show the main menu, student registration and login, quiz attempt, score calculation, result history, administrator question management, and the automated test.

11. Challenges Faced

Input validation, persistent file storage, modular class design, and correct score calculation are key implementation challenges.

12. Learnings and Key Takeaways

The project demonstrates object-oriented design, inheritance, polymorphism, collections, file handling, exception handling, validation, and command-line application structure.

13. Future Enhancements

Database integration, password hashing, multiple quiz categories, random questions, timed quizzes, leaderboards, and a web or graphical interface.

14. References

Oracle Java Documentation: <https://docs.oracle.com/en/java/> . Java platform documentation and course materials.